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An IoT-driven machine learning system for real-time smart crop recommendation and optimization in precision agriculture

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Abstract

Agriculture is essential to global food security and sustainable development. The use of machine learning (ML) and Internet of Things (IoT) has the potential to enhance crop production and improve resource management through enhanced precision agriculture. This article describes a smart crop recommender system developed to support both ML-based decision making and real-time environmental sensing. At the core of the system is a capacitive soil moisture sensor and a temperature-humidity sensor interfaced using an Arduino Nano. Using a nRF24L01 transceiver employing the 2.4 GHz Enhanced Shock Burst (ESB) protocol, the sensor information can be wirelessly transmitted. Data collected from the sensors are sent to an ESP32 module that posts the information to the Blynk web application through Wi-Fi, allowing real-time remote access. As the data source for the crop recommender model, the Blynk application provides the data needed to perform data analysis on the various ML algorithms (Random Forest, Bagging, Decision Tree, Gradient Boosting, and Enhanced Gaussian Naive Bayes [EGNB]) to determine the most suitable type of crop based on the current soil and environmental conditions. Based on the analysis, the EGNB model showed superiority with an accuracy of 99.55% with respect to precision, recall, and F1-score. This system provides an economically viable and expandable approach to smart agricultural practices, which integrates cutting-edge machine-learning modelling with digital tools for obtaining data in real-time (IoT). These capabilities allow users to receive and utilize actionable, data-driven guidance in real time when making decisions about crop production.

Keywords Crop recommendation system, Internet of things, NRF24L01, Enhanced Shockburst (ESB), ESP32, Blynk app, Machine learning, Enhanced Gaussian Naive Bayes, Precision agriculture, Real-time monitoring, Wireless sensor networks

1 Introduction

For centuries, the Indian economy has relied heavily on Agriculture; it employs more than half of India's population, and its contributions to GDP are large. During the past few years, rapid changes in agriculture have created many issues that hinder farmer profitability. These issues include fragmented land-holdings, nutrient depletion of soils,



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drought conditions caused by erratic monsoons, limited irrigation or water supply, and increases in crop losses due to climate change. Additionally, to these issues, farmers do not have adequate access to timely agronomic advice, and rely on outdated agricultural methods/production systems. As a result, these barriers create a situation of poor crop-planning, over use of fertilizers, and crops that often yield less than their cost-of-production. According to the Indian Council of Agricultural Research, one of the most significant contributors to decreased agricultural output and lower farm income in the different climate zones in India is poor crop rotation. Because of this, there is now an increasing demand for intelligent, localized, and sustainable tools that provide farmers with the ability to make evidence-based choices about their farms. One such promising opportunity to assist farmers in this regard is Precision Agriculture (PA). It is an agronomy method that uses a combination of advanced technologies, including sensors, GPS, Satellite images, and Drones, along with AI and ML, to help farmers understand the variability that exists within a crop and utilize their inputs (water, fertiliser, pesticides, etc.) more effectively. The purpose of PA is to allow farmers to produce increased crop yields using fewer inputs and with reduced environmental impact. Over the last ten years, the use of PA approaches has started gaining traction in India, with pilot projects being implemented in Punjab, Maharashtra, Karnataka, and Tamil Nadu. Despite progress, the adoption of PA from small and marginal farmers is still limited because of high costs, limited knowledge of the new advances, and low connectivity within rural areas.

The Crop Recommendation System (CRS) of PA will be an important component of the exploration into PA. The CRS provides a scientifically based recommendation on the most suitable crop for the field based on environmental and soil data. Crop recommendations prior to the implementation of PA were mainly circulated based on large agro-climatic zones, without considering the micro-level variations in soil nutrients and varying weather events. Consequently, farmers would choose crops without consideration for their crop-soil compatibility or the best soil-nutrient combinations, often resulting in poor crop performance and financial loss. Through the use of IoT and ML, it is now possible to develop a real-time, site-specific CRS that transforms sensor-generated information from the field into useful information. Many research studies have demonstrated that such systems are effective. Gottemukkala et al. [1] developed a supervised machine learning-based model using soil N-P-K data. Senapaty et al. [2] developed the IoTSNA-CR framework to make an intelligent crop recommendation after developing an algorithm with cloud, sensors, and classification. These mostly remain at the academic level or geographically bounded examples. There is a gap for a generalizable, scalable, and accessible solution that combines hardware, uses cloud machine learning, and has user-friendly tabs.

In this sense, we offer an end-to-end IoT crop recommendation system as an example that can be implemented in practice. Our crop recommendation system is based on the features and a machine learning engine trained on a validated crop dataset of 2200 samples with parameters include nitrogen, phosphorus, potassium, pH, temperature, humidity, rainfall, and best crop label. For the machine learning engine, we used several classifiers: Random Forest, Gradient Boosting, Bagging, Decision Tree, and our proposed Enhanced Gaussian Naïve Bayes (EGNB), using normalized and non-revealing input features. In terms of hardware, we created a low-cost prototype using an Arduino Nano with a capacitive soil moisture and temperature-humidity sensor. We used the

nRF24L01 module with Enhanced ShockBurst (ESB) protocol for wireless data communication to an ESP32 receiver, and then transferred it to the Blynk web app. This data is now used in real-time by our trained ML model to provide crop recommendations. These recommendations were represented in a straightforward and symbolic interface on the mobile/web application that was designed with farmers who have limited digital literacy in mind. Our work is original in being able to combine all of these real-time environmental sensing capabilities, low-cost open-source hardware technology, and highly accurate ML predictions into one convenient and usable interface. This integrated solution fills critical gaps in the PA infrastructure for farmers in India, as well as offers a path forward to scalable and low-cost data-driven solutions to equip farmers to take more control over their work to secure national food sustainability.

The rest of the article is structured as follows: Sect. 2, Literature Review, presents an overview of previously published research on the convergence of machine learning (ML) and the Internet of Things (IoT) for agricultural decision-making. Section 3, Materials and Methods, outlines the methodologies and materials employed in the study, including IoT components and sensors, data transmission and analysis techniques, machine learning methods, and the overall system framework. In Sect. 4, Experimental Results, presents a comprehensive evaluation of the proposed IoT-ML based crop recommendation system. Section 5, Conclusion and Future Scope, summarizes the key findings of this study and future research directions.

2 Literature review

Over the last few years, we've seen a paradigm shift in how farmers and other decision-makers in the agriculture industry make decisions using data. Historically, farmers relied almost exclusively on past practices and the knowledge of those around them to make farming decisions, but now with the advancement of ML systems using IoT technology, farmers can make use of real-time data to create precise, data-driven solutions for their farms. Developing ML and IoT technologies has been the focus of much of the recent agricultural research, resulting in the development of various systems for crop selection, soil administration, irrigation management, and yield maximization (See Table 1).

Islam et al. [3]. The authors incorporated sensors into the advisory platform for the FC-28 (moisture sensor), DHT11 (temperature), and JXBS-3001 (for estimating nutrients NPK). This platform was capable of producing the live data fields necessary to be processed by using machine learning models such as random forest, logistic regression, LightGBM, and neural networks; thus, achieving not only high accuracy in crop predictions during field tests, but also enhanced advice on managing fertilizer use (an essential element of sustainable agricultural practices). Additionally, through the incorporation of a mobile app that provided incentives to consumers who evaluated the crop quality, they extended the research beyond simple crop production and developed an end-to-end agri-ecosystem incorporating all aspects of the food production chain. Finally, their research shows that real-time sensing combined with artificial intelligence can transform long-standing paradigms of time-intensive agricultural operations into more efficient and responsive ones and also create opportunities for greater sustainability.

After recognizing the need for integrated advisory systems Gottemukkala et al. [1] presented a comprehensive crop-and-fertilizer recommendation model for resource constrained settings like Rwanda, but could have relevance in other areas with diverse

Table 1 Literature review of the few studies which uses ML or IOT for for agricultural decision-making

Refer- ences no.	Models used	Model accuracy	Features used
[1]	Random forest, logistic regression, LGBM, neural networks	Not specified (Field validated)	Soil monitoring, crop advisory, fertilizer management
[2]	Neural networks, rule-based	97%	Crop and fertilizer recommendation
[3]	Decision trees, KNN, random forest, neural networks	91%	Crop recommendation, rainfall prediction
[4]	IoT + Arduino system (no ML model)	Real-time irrigation control (no accuracy stated)	Smart irrigation management
[2]	Logistic regression, ridge regression, decision trees, XGBoost, LightGBM	96%	Crop recommendation with alerts
[3]	Optical, electrochemical, conductivity-based sensors (Review)	Not applicable (sensor review)	Review of soil nutrient detection technologies
[5]	Decision tree, KNN, random forest, gradient boosting, SGDC	100% with SGDC after SMOTE	Big data-based personalized crop recommendation
[6]	Optical sensor system (no ML)	Classification into High/Medium/Low (not %)	Soil NPK detection
[7]	SVM with GLCM feature extraction	Good accuracy (exact % not given)	NPK detection + IoT irrigation
[8]	Random forest (RF) + SHAP + LIME	High (exact % not given)	Interpretable crop recommendation
[9]	Naïve Bayes	85.71%	Food crop planting recommendation
[10]	Random forest + extreme gradient boosting (RFXGB) + DeepFM	Superior performance (no exact %)	Crop and fertilizer yield recommendation
[11]	Gradient boosting machine (GBM)	Better than traditional ML (no exact %)	Personalized sustainable farming guidance
[12]	K-Nearest neighbours (KNN)	High accuracy (not quantified)	Crop recommendation based on current soil & weather data

agro-ecological systems. Their model utilized neural networks for the prediction of the crop, and a rule-based engine for the fertilization recommendation thus creating a hybrid model. With the addition of cloud-connected IoT sensors the model reached an accuracy of over 97%, which is very good considering the accuracy levels necessary in a real-world deployment context. The study integrated pre-sowing and post-sowing crop care, along with a nutrient planning module, in a single system, which is at the forefront and offers a global vision. They author also highlighted the need for original integration, correlation analyses, historical data, and modularity of systems, are important elements in order for them to be relevant in complex agro-ecological systems and smallholder farmer systems. The researchers also presented real-time systems in order to provide their ‘real-time’ learning, but also spoke to sustainable fertilizer use, data integrity, and adaptable systems.

Another significant contribution to decision support systems is by Doshi et al. [4] on the Agro Consultant system, a multi-label classification model that breaks the mold of a single component because it incorporates soil and environmental measures simultaneously. The Agro Consultant model included pH, aquifer depth, temperature, rain and location data, making it an improvement over modelling only soil or climate data. In addition, they tested everything from Decision Trees, K-Nearest Neighbors (KNN), Random Forest, and Neural Networks (the Neural Network had the best classification accuracy (91%)) from a wide range of machine learning methods. A valid and exciting

addition that the authors presented in their study was the thinking about second rain-fall predictions to make the crop suggestions a bit better. Similarly, by including their special “map” interface in the model (Fig. 5), they allowed the user to spatially consider crop suggestions and ascertain where markets may be vulnerable to overproduction and saturation, subsequently directing them toward economical sustainability. They demonstrated the utility of recommended actions compounded by different factors and spatial analytics.

Singh and Saikia [5] examined smart irrigation management from the perspective of optimizing agricultural resources by studying the components that are interconnected in the crop recommendation system. They developed a DIY, low-cost Arduino-based irrigation control system that incorporates soil moisture, temperature, water flow and ESP8266 WiFi modules to allow farmers to monitor their water usage via a web dashboard and control their irrigation system remotely. This control system provides an important tool for minimizing water usage in a country such as India where the majority of the land is suffering from prolonged drought. One of the reasons that the authors believe their Control system is unique compared to other commercially available systems, is that the Control system is designed as a low-cost usable product for the small-holder farmer, who would likely not be able to purchase high-specification or hi-tech type systems. The Control system is an example of how Internet of Things (IoT) technologies can offer direct solutions to address issues of resource scarcity, specifically addressing one of the main pain points for both Indian and global agriculture.

Recently, Kandulata et al. [6] put forth a robust, ensemble-approach to crop recommendation system based on classifiers for Logistic Regression, Ridge Regression, Decision Trees, XGBoost and LightGBM and trained it using a data set consisting of Rainfall Temperature, Land-Area and District-Level Metadata [5]. This platform was able to develop a crop recommendation accuracy of 96% and possessed a real-time alert (SMS and Email) to their users. The ensembles utilized in the study, with hyper-parameter tuning, provided stability and flexibility of the predictions. The incorporation of a real-time advisory system and communication channels to users demonstrate the capability to employ this type of system in rural and semi-urban farm areas. Their research clearly illustrates how integrating algorithmic depth and reliable delivery mechanisms (mobile) can provide a greater number of producers the opportunity to access and take action on precision agriculture.

Dattatreya et al. [7] have published a comprehensive review of the current approaches toward developing technology that will allow for the incorporation of soil nutrients into crop management systems that are supported by artificial intelligence (AI). Their review discusses a wide variety of methods including traditional laboratory tests and modern high-tech sensors, including reflective optics, laser-induced breakdown spectroscopy, and electrochemical sensing devices. One major area where they identified evolving trends in sensor technology was the trend towards non-intrusive, portable, and real-time soil nutrient monitoring devices, reflecting and supporting the possibility of the use of MEMS technology, polymeric and nanomaterial-based sensors in the future. They further emphasized the importance of a more continuously updated soil and climate databases that incorporates historical data and real-time conditions. The K-Nearest Neighbor (KNN) algorithm was used for recommendation purposes, based on real-time moisture, temperature, rainfall, and pH, with high accuracy and usability in operation.

The authors found that improving the consistency and quality of sensor data, as well as allowing for flexible and tunable algorithms, will be essential to support the development of new-generation Agriculture Decision Support Systems.

These studies highlight an increasing consensus: modern agriculture must be reliant on intelligent, sensor-enabled, real-time decision-making systems. While many of these “systems” have a high level of accuracy and field applicability, very few are fully integrated from sensing to user output and can deliver that output in a cost-effective, scalable technical solution accessible by farmers in some of the developing communities like those in India. The present study builds upon this observation and hopes to address this gap by developing a completely integrated crop recommendation system with real-time sensing, upgraded machine learning models, cloud deployment, and a simple and usable interface for rural users. This study extends the originality and perspective provided by existing literature as a practical, research-conscious contribution to the field of smart and sustainable agriculture.

3 Methods and materials

This section outlines the methodologies and materials employed in the study, including IoT components and sensors (3.1), data transmission and analysis techniques (3.2), machine learning methods (3.3), and the overall system framework (3.4).

The hardware components were procured from Robu.in, a reputable Indian electronics and prototyping supplier. Key components included the Arduino Nano, developed by Arduino.cc (Italy), for data acquisition, and the ESP32, developed by Espressif Systems (China), for wireless data transmission and cloud connectivity. Environmental sensing was achieved using the DS18B20 temperature sensor (Maxim Integrated, USA) and the Capacitive Soil Moisture Sensor v1.0 (manufactured by multiple Chinese vendors). Wireless communication was enabled by nRF24L01 transceivers, developed by Nordic Semiconductor (Norway), operating at the 2.4 GHz frequency [8].

3.1 IoT components and sensors

This study used three types of sensors—Capacitive Soil Moisture Sensor v1.0, DS18B20, and an NPK sensor—integrated with the Arduino Nano and ESP32 microcontrollers. Wireless communication was established using nRF24L01 + transceivers on the 2.4 GHz frequency band. A circuit diagram is shown in Fig. 1, and the component specifications are summarized in Table 2.

3.1.1 Capacitive soil moisture sensor v1.0

Soil moisture content is determined by measuring the volume of water in the soil using this sensor. Capacitive sensors, as opposed to their resistive counterparts, are not susceptible to corrosion due to prolonged usage; therefore, they are much more reliable for consistent soil-monitoring applications, due to their ability to respond to fluctuations in moisture by measuring the dielectric characteristics of the affected soil.

3.1.2 DS18B20 temperature sensor

Measuring soil temperature is an important component in the growing process as it helps dictate seed germination, nutrient absorption, soil properties (microbial activity), and plant health. Therefore, by keeping soil at appropriate thermal levels, farmers

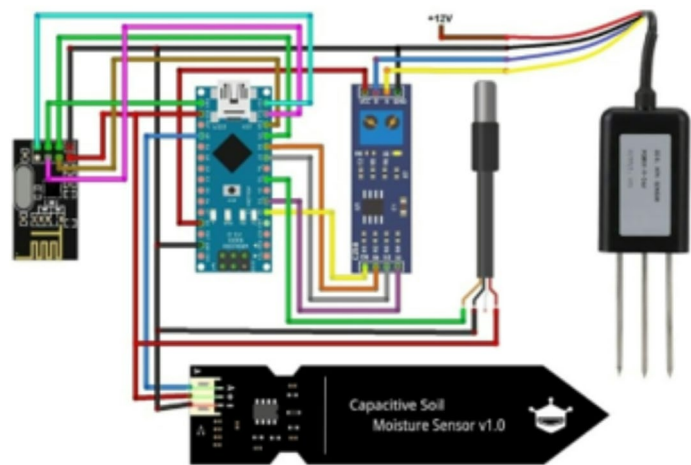


Fig. 1 Circuit diagram of the proposed system Transmitter side

Table 2 Configurations and specifications of the proposed IoT system components

Component	Specifications
Arduino Nano	Microcontroller board based on ATmega328P. Operates at 5 V with a clock speed of 16 MHz. Offers 22 I/O pins (14 digital, 8 analog). Handles sensor data acquisition and wireless transmission via nRF24L01.
ESP32 (NodeMCU)	Wi-Fi and bluetooth-enabled microcontroller by Espressif Systems. Features a dual-core processor, operates at 3.3 V logic, has multiple GPIOs. Receives sensor data and uploads it to Blynk cloud via Wi-Fi.
Capacitive soil moisture sensor v1.0	Non-corrosive sensor for measuring soil moisture. Operates between 3.3–5 V. Outputs analog voltage corresponding to moisture level.
DS18B20 temperature sensor	Waterproof digital sensor with 1-Wire interface. Measures – 55 °C to + 125 °C with ± 0.5 °C accuracy. Requires 4.7 kΩ pull-up resistor on data line.
NPK sensor	Measures nitrogen, phosphorus, and potassium in soil. Operates on 5–9 V DC. Communication via UART or RS485. Measurement range: 0–1999 mg/kg.
nRF24L01 + module	2.4 GHz wireless transceiver by Nordic Semiconductor. Operates at 3.3 V with SPI communication. Supports Enhanced ShockBurst™ for low-power, high-speed data transmission. Used between Arduino Nano and ESP32.

decrease the risk of both frost and heat stress affecting their crops. The DS18B20 sensor records the soil temperature and allows farmers to maintain ideal conditions for their crops by providing accurate temperature readings.

3.1.3 NPK sensor

Through soil testing, this device can help farmers identify the amount of nitrogen, phosphorus, and potassium present within the soil. These three elements are essential macronutrients that support healthy plant growth and thus increase overall crop production through proper management of soil fertility and precision application of fertilization methods.

3.1.4 Arduino nano

An Arduino Nano is used as the main microcontroller which collects readings from the NPK Sensor, Soil Moisture Sensor, and Temperature Sensor and sends the data through a wireless module after preprocessing. Due to its small size and ability to connect with multiple sensors, it is well-suited for Precision Agriculture at a low cost.

3.1.5 NodeMCU (ESP32)

Based on its low cost and low power consumption for supporting integrated Wi-Fi and Bluetooth, the ESP32 Microcontroller provides Users with the ability for real time data transfers to the Cloud, thanks to its Dual Processor architecture and a number of GPIO Ports available on the device. Additionally, the ESP32 will act as the end user receiver and as the Connection Point for Internet-based services within this Architecture.

3.1.6 nRF24L01 wireless module

Wireless data is transmitted with the help of an nRF24L01 module. It connects the field sensors with the central data processor to provide a low latency link to allow reliable communications. The nRF24L01 operates in the 2.4 GHz band, which makes it possible to wirelessly transmit sensor data without delay. It sends the sensor information to the ESP32 that forwards it to the cloud for archival. The data is collected automatically so there is no operational downtime and allows real-time soil monitoring (Fig. 2).

3.2 Data transmission using enhanced shockburst (ESB) protocol

The communication module for the relay node in this work utilizes an enhanced shock burst (ESB) protocol, which is supported by the nRF24L01 + transceiver, to allow for a seamless transition between wireless communication and an efficient and reliable way to send data from the sensor nodes to the computer system. The ESB is ideal for this system because it provides an efficient way to transmit data over long distances and supports very low power usage, supports automatic acknowledgement (ACK) and retransmission, and is an effective way to guarantee that data is sent reliably, even in hostile environments like agriculture. The proposed method involves using the Arduino Nano as a microcontroller to collect environmental variables (including soil moisture and temperature) and sending those values to the nRF24L01 + module via the 2.4 GHz ISM band in structured packets, along with time stamps and node identification numbers [10].

The transmitted data is received by an ESP32 microcontroller with an nRF24L01 + Module. The auto-retry function of the ESB Protocol guarantees a high level of reliability and low potential for data loss through Acknowledgment. When the ESP32 receives the sensor information, it sends it to Blynk via Wi-Fi. Blynk serves as the connection point between the hardware and software levels of the system, enabling the user to monitor the real-time readings of all the sensors and to obtain all the data for its Crop Recommendation Website Web Application, where the Back-End will access all the Sensor Data collected by Blynk and utilize Machine Learning Algorithms to identify which Crop(s)

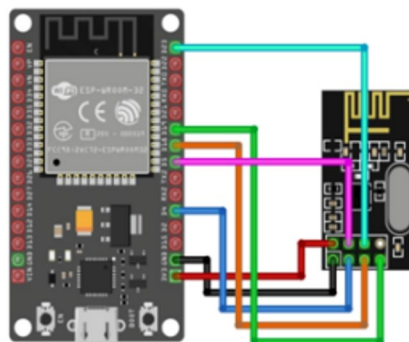


Fig. 2 Circuit diagram of the proposed system Receiver side

would be best suited for the current environmental conditions (as measured by sensors), and thus, recommend the appropriate Crop(s) to plant. Due to the low latency, energy efficiency, and high reliability associated with the ESB Protocol combined with Blynk's extremely user-friendly Cloud Interface, the user is able to have a complete open-ended and scalable data pipeline, from the Sensor Level in the Field to the Intelligent Machine Learning-Based Crop Recommendation. Data preprocessing (e.g., filtering and formatting) is performed prior to feeding the data into the Machine Learning Algorithms to ensure both accuracy and compatibility [11].

3.3 Machine learning methods

The following segment pertains to the integration of machine-learning algorithms within an analysis pipeline for processing sensor data and yielding insightful information regarding soil nutrient levels and crop advisement. Five separate classification algorithms were utilized in this study; Enhanced Gaussian Naive Bayes (EGNB); Random Forest Classifier; Bagging Classifier; Decision Tree Classifier; Gradient Boosting Classifier. Training and testing these models allow the identification of optimal crops based upon real-time environmental conditions.

3.3.1 *Enhanced Gaussian Naive Bayes (EGNB)*

The enhanced model was designed to improve the classical Naive Bayes (NB) algorithm without needing the strict probability distributions that characterize advanced NB models (e.g., multinomial NB). To provide adequate scaling of the input features in order to model them appropriately within the constraints of the standard NB model framework, the scikit-learn version uses a StandardScaler (which assumes the input features are distributed normally) which allows for better estimates of likelihood for all continuous input features across different samples. In order to reduce numerical instability during likelihood estimation, the smoothing factor of $\text{var smoothing} = 1\text{e-}8$ was applied. To create a complete model, the model was assembled into a two-phase process, namely: the model performs standardization on the input features it receives from the users; subsequently it classifies those standardized features after estimating their likelihoods. By doing this, the input features are standardized against one another and therefore can be handled on similar scales to each other and are able to significantly improve classification accuracy compared to using the features individually.

3.3.2 *Random forest classifier*

An ensemble learning algorithm known as the Random Forest Classifier creates many decision trees throughout training and represents the final classifications as the class that is most frequently predicted by each tree. The introduction of randomness into both feature selection and training sample (bootstrap aggregating) during the building of decision trees helps reduce overfitting and result in a more generalized, robust model. The Random Forest Classifier was created with the intent of reducing overfitting while maintaining a more generalized model. For this research, the Random Forest Classifier was created using scikit-learn [13] with 100 estimators (the default setting) and uses the Gini impurity metric to evaluate and select the best choice based upon several different possible selections. The output from the trained Random Forest model provides valuable insight into the relative importance of the various measured features used within the

study, including soil and environmental variables (e.g., temperature, moisture, and NPK levels) used to predict the response of different crops [14].

3.3.3 Bagging classifier

The Bagging Classifier is another ensemble learning method that reduces variance by constructing various base learners on various random sub-samples of the dataset and then taking the mean of the output of each base learner. A Bagging Classifier has been created with Decision Trees being the base estimators [15]. By creating bagged datasets from a variety of sources, Bagging Classifiers are more stable than the individual base learners and reduce the likelihood of overfitting, allowing better management of high-variance characteristics (e.g., rainfall, moisture level) and increased tolerance to noise and volatility when operated in real-time systems such as sensors.

3.3.4 Decision tree classifier

A Decision Tree Classifier is a supervised learning algorithm that partitioning data (a supervised learning algorithm) by using the values of features recursively by branches until a decision is achieved. The Decision Tree Classifier offers both interpretability and low computational complexity. In this paper, a Decision Tree Classifier was constructed through the use of the Scikit-learn library with a Gini index as the splitting criterion. To improve the quality of the model generated, pruning techniques were utilized. As this model was substantially less complicated than previously described ensemble techniques, it represented a benchmark for further comparison [16].

3.3.5 Gradient boosting classifier

Gradient Boosting is a strong method for generating an ensemble of classifiers, in that it tends to create models through the sequential application of new classifiers, so that they correct the errors made by all previously built classifiers. Gradient Boosting uses weak classifiers (typically decision trees) and optimizes their combined performance by finding the best value of a loss function using gradient descent techniques. The Gradient Boost Classifier was utilized, with a fixed number of estimators and hyper-parameter setting for the learning rate [17]. The techniques learned in this research, including tuning learned rates, yielded improved predictive accuracies by specifically focusing on the most difficult-to-classify samples (instances) and by reducing bias and variance in the model. This approach was effective in many difficult socio-environmental scenarios where non-linear interactions exist between classifier features.

3.4 Proposed framework

This framework presents an affordable and scalable IoT-based Smart Agriculture system featuring machine learning methods to support precision agriculture and intelligent crop recommendations. Data from the environment is collected via a ESF-Nano Temperature Sensor and DS18B20 Temperature Sensor via a Capacitive Soil Moisture Sonar (used to collect data on soil moisture level) & DS18B20 Temperature Sensor connected to an Arduino Nano Micro-Controller. These sensors will provide a means of monitoring the soil moisture level and temperature as they change over time. Collected sensor data will then be transmitted from the Arduino Nano to the ESP32 Microcontroller via the nNRF24L01 Wireless Transceiver using the ESB Protocol (Enhanced Shock Burst)

operating at 2.4 GHz, providing low latency and low power consumption. This ESP32 will serve as the Central Node, collecting the sensor readings from each sensor and passing them up to the Blynk IoT platform over Wi-Fi, where users can continuously monitor and view live data from the sensors. Collected Sensor Data is stored in the Blynk Database and is made available for Machine Learning Algorithms to be either computed in the cloud or on premise.

The models for classification were trained and tested on labeled crop datasets, and include: Random Forest, Bagging, Decision Tree, Gradient Boosting, and Enhanced Gaussian Naive Bayes (EGNB). All of the models use environmental parameters to make crop recommendations based on the environment. As a result, EGNB had the highest accuracy, at 99.55%, compared to all other algorithms for precision, recall, and F1-score. The entire solution combines real-time IoT-based sensing, wireless transfer of data to the cloud, analytic intelligence (ML) tools to enable data-driven decision-making and improved crop production [18].

To determine how well the dataset would generalize, a stratified split was performed. This resulted in a training dataset made up of 80% of the overall dataset, and a separate, independent test dataset composed of 20% of the dataset when evaluated using this method. A “stratified” approach ensures that the training and test sets maintain the same proportions of each crop type in the original dataset, thereby minimizing the potential for class imbalance bias to affect the evaluation of model performance. The input features represented in the dataset are a diverse range of heterogeneous variables, which include nitrogen (N), phosphorus (P), and potassium (K) content concentration parts per million (ppm), temperature (degrees C), relative humidity (%), amount of rainfall (mm), and soil pH. One of the major preprocessing steps needed in order to be able to compare these heterogeneous input features is called feature scaling. Feature scaling uses a “standard-scale” approach to apply a mean of 0 and standard deviation of 1 to all features in the training dataset. This is the same approach used when applying the normalization steps in probabilistic classifiers (e.g., Gaussian Naive Bayes) because these classifiers assume that their input data are normally distributed. Feature scaling provides the model with numerical stability by preventing any one variable from dominating the learning process due to the difference in the magnitude of its numerical value compared to other variables in the dataset. To prevent data leakage, the same feature scaling criteria used to produce the training set were used to produce the test set.

To improve stability and decrease the amount of variation caused by only using one data split, the authors decided to use 5-fold cross-validation while building their model. Each training dataset was divided into five equal pieces; four were used to train the model, while one was set aside for testing each cycle of training. This was done five times so that eventually, every piece of the training dataset would be tested once. After all, five cycles had been completed, a summary of all the results (averages) would indicate how consistently well the model had performed throughout the testing. Because of the significant difference in climatic conditions from year to year within agricultural datasets, the authors determined that using the 5-fold technique would be important for determining stability or performance consistency across training and test datasets.

Final selection of machine learning algorithms was made based on predictive accuracy, interpretability, and computational speed to be used in Precision Agriculture IoT Products. The algorithms were divided into tree-based ensembles and statistical models.

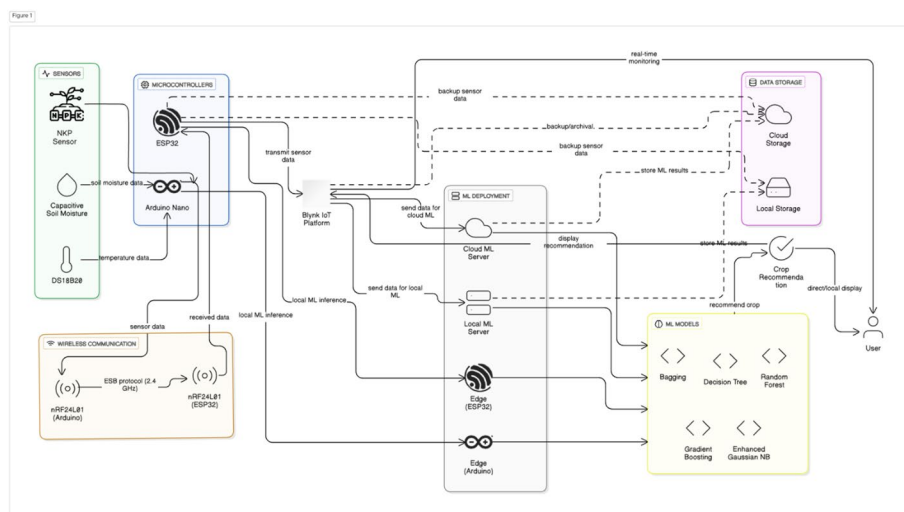
The tree-based ensemble techniques selected included Random Forest, Bagging and Gradient Boosting, which capture nonlinear relationships between variables, handle interactions between features, and increase robustness to variances in input datasets. Additionally, these algorithms provide excellent base models for comparison when creating algorithms to support decisions in an agricultural setting (Fig. 3).

4 Experimental results

This section presents a comprehensive evaluation of the proposed IoT-ML based crop recommendation system. It is structured into five key subsections to facilitate clarity and detail. Section 4.1 outlines the hardware and network configuration deployed in the field. Section 4.2 describes the methods adopted for data acquisition, cleaning, and pre-processing. Section 4.3 focuses on the evaluation metrics used to quantify the system's classification performance. Section 4.4 discusses and interprets the experimental outcomes obtained from various machine learning models. Finally, Sect. 4.5 introduces the complete integrated IoT and ML-enabled platform designed to support smart agriculture through real-time monitoring and predictive analytics.

4.1 Field deployment and experimental setup

Equipment consisting of a soil-sensing unit and one Gateway Unit using low-cost, low-power hardware components. The Sensing Unit has a capacitive soil moisture sensor and a DS18B20 digital temperature Sensor connected to an Arduino Nano microcontroller. The Capacitive Soil Moisture Sensor tracks the level of moisture in the soil, while the DS18B20 digital temperature sensor tracks the temperature of the soil. Both sensors continuously monitored soil moisture and temperature, providing real-time environmental conditions all necessary for crop prediction. Sensor data was submitted using an nRF24L01 transceiver that operated on the 2.4 GHz ISM band and utilized the Enhanced ShockBurst™ (ESB) protocol [19], which provides prompt point-to-point transmission that is reliable and low power. The receiving unit was built using an ESP32 microcontroller to act as the gateway. The ESP32 received the wireless data packets from



eraser

Fig. 3 Circuit diagram of the proposed system Transmitter side

the remote Arduino-based node and forwarded them to the Blynk IoT cloud platform via Wi-Fi. Blynk was the cloud interface that allowed real-time visualization and remote monitoring via the smartphone application or web dashboard. With this structure, low-power long-range communication was possible with ongoing cloud access to the sensor data [20]. For simulation of soil fertility conditions, additional synthetic features representing NPK (nitrogen, phosphorus, and potassium measured in ppm) values were created to supplement the original dataset so that the proposed approach would have more complete soil health profiles during model training and crop recommendation. The entire system was tested under open field conditions with varying moisture, temperature, and nutrient levels to evaluate the system responsiveness, reliability of transmission, and robustness of the proposed framework [21].

4.2 Data collection and preprocessing

The Crop Recommendation Dataset includes 2200 observations that have 8 features to help aid decision-making in agriculture by identifying the right crops to grow based on soil and climatic conditions. Each row includes data collected on soil nutrients, nitrogen (0–140 mg/kg), phosphorus (5–145 mg/kg), and potassium (5–205 mg/kg), that are important to plant functions such as leaf growth, root development, and water regulation. Climate variables include temperature (8.82–43.67 °C), humidity (14.25%–99.98%) and rainfall (20.21–298.56 mm/year) that represent the suitability of crops during those conditions [22]. Soil pH ranges in acidity from 3.5 to 9.93, affecting nutrient availability and compatibility of crops. For example, tea grows in acidic soil, while barley likes alkali soil. Temperature affects growing seasons of crops; for example, wheat does better in cool climates, while rice prefers warmer climates. Humidity and precipitation also influence growing conditions for crops such as drought-tolerant millets and maize, which need less water, compared to growing rice, sugar cane, etc. Target crop categories include cereals (rice, wheat, maize), legumes (chickpea, lentil), vegetables (tomato, onion, potato), fruits (banana, mango, watermelon) and industrial crops (cotton, jute, coffee). The data set provides various combinations of soil fertility and climatic conditions for a multitude of crop types applicable to machine learning in precision agriculture. This database will provide farmers with improved and accurate crops to grow based on their specific fertilizer, irrigation, and yield efficiency, as well as overall sustainable agriculture.

4.3 Data exploratory analysis

This data slice has five samples identified as “Rice,” with each having values for seven essential agronomic and environmental factors identified as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH and rainfall. Nitrogen values vary from 60 to 90 ppm (avg. 77 ppm), which supports vegetative growth, while phosphorus levels (35–58 ppm) support root development early on. Potassium values (40–44 ppm) help to maintain moisture balance in plant and aid the plant in withstanding diseases, both of these are important for rice during grain filling. Given this, the values of NPK suggest soils with moderately fertile conditions support adequate nutrient levels for rice. The temperature values had variations of 20.15 °C to 26.49 °C with average of a 22.85 °C, which is ideal for rice, as rice grows best in temperatures between 20 and 30 °C. The humidity values in West Bengal and Assam ranged from 80.15 to 82.32%, showing rice

can live in moisture-rich soils, which is important for regions prone to the monsoons uses plentiful moisture retaining areas. The pH values of the sample soils varied from 6.5 to 7.84, that most ranged from slightly acidic to neutral, and may need slight correction to aid rice where repeated cycles of alkaline-prone soils occur in their seasons. The rainfall ranges from 202.93 to 263.96 mm with average of a 240 mm—which is completely within the water requirements of rice. Overall, there were uniform and optimal environmental conditions, which simply supports cultivation of rice [6]. The values in this data can assist machine learning models to identify typical growing conditions when growing rice, and have value for improving crop recommendations and provide discovery for intelligent and knowledge-based, sensor-based, agricultural systems.

The above image (Fig. 4) illustrates the distribution of key features in the dataset, including nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. These distributions provide insights into data variability, potential skewness, and the need for preprocessing steps such as normalization or transformation before applying machine learning models.

Table 2 provides the average nutrient needs of 22 crops (in kg/ha), pH, and growing conditions. The average nutrient requirements for an example crop banana is about 100.23 kg of nitrogen, 82.01 kg of phosphorus, and 50.05 kg of potassium per ha. The pH for banana production occurs at pH 5.98, which is quite low on the pH scale—that is, mildly acidic. For example, mungbeans need 20.99 kg of nitrogen, 47.28 kg of phosphorus, and 19.87 kg of potassium at pH 6.72 (acidic to neutral) soils. Likewise, maize, which is widely cultivated with the pH conditions of 6.25 (acidic), grew well with 77.76 kg of nitrogen, 48.44 kg of phosphorus, and 19.79 kg of potassium [23].

In contrast, cotton has a fair bit of a need of nitrogen with up to 117.77 kg N, at moderate to average amounts of phosphorus (46.24 kg) and potassium (19.56 kg) needs, with a pH of 6.91, which is neutral. Rice, typically used as a staple crop, needs about 79.89 kg of nitrogen, 47.58 kg of phosphorus and 39.87 kg of potassium at soil pH 6.43 from mildly acidic soils. Similarly, other crops include apple and grapes, which indicate value

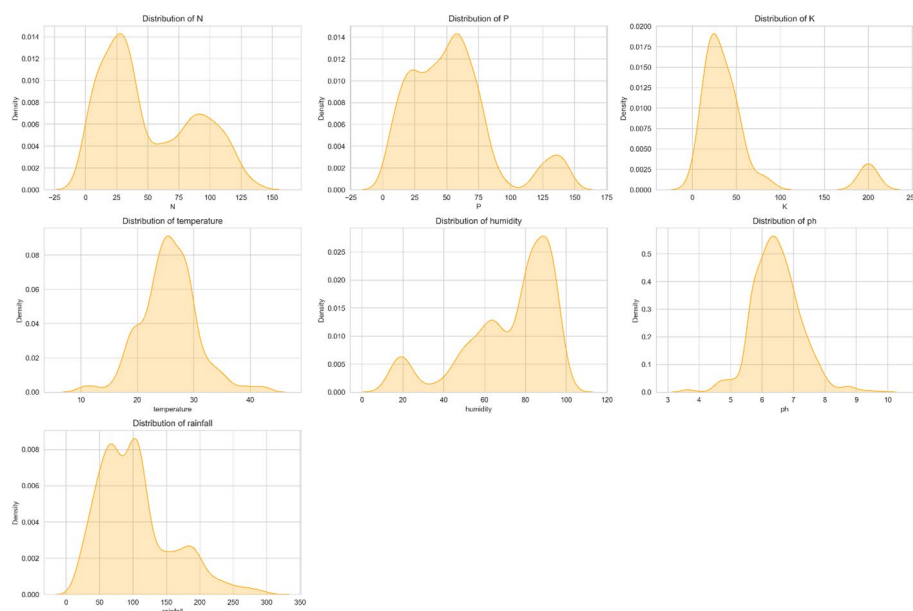


Fig. 4 Kernel density estimation (KDE) plots r side

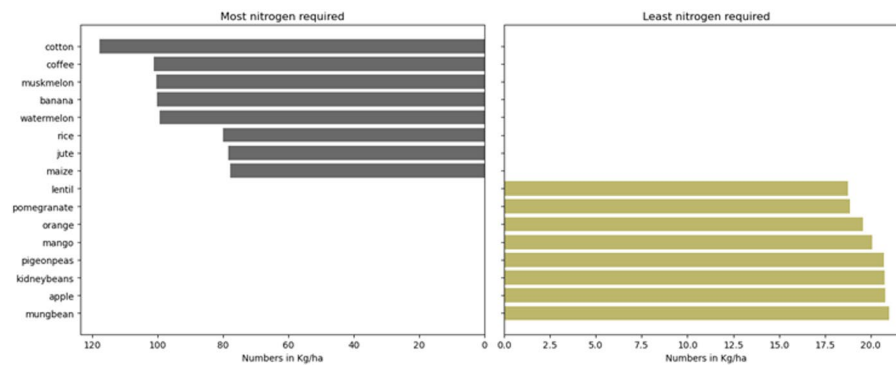


Fig. 5 Analysis regarding most and least nitrogen-requiring crops

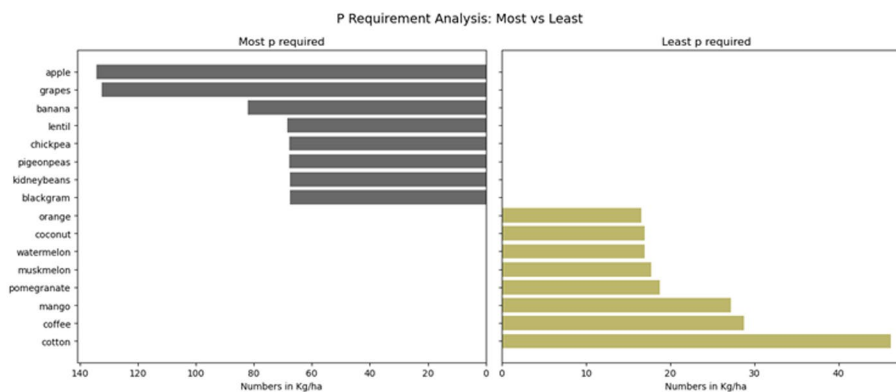


Fig. 6 Analysis regarding most and least phosphorus-requiring crops

indicative of potassium (nearly 200 kg/ha) and phosphorus, with a relatively mild acidic value of pH 5.93 (apple) and 6.03 (grapes).

Figure 5 shows Comparative analysis of nitrogen requirements of crops (in Kg/ha). The left corner highlights that crops with the most nitrogen requirement which shows cotton has the most nitrogen requirement followed by coffee, muskmelon, and banana. The right presents crops which have the lowest nitrogen requirement which were mungbean, apple, kidney beans, and pigeonpeas. The analysis of nitrogen requirements specifies which crops require more nitrogen and which crops efficiently use nitrogen [2]. This assessment is essential for fertilizer use in a sustainable manner and will indicate which crop you could grow depending on the nitrogen that is available.

Figure 6 shows insight that the Crops were compared for phosphorus (P) requirements (Kg/ha). The left-hand panel lists the crops with the highest P requirements, including apple and grapes, followed by banana, lentil and chickpea [2]. The right-hand panel includes the crops with the least amount of P requirements, including cotton, coffee, mango and pomegranate. This in-conjunction analysis provides information on crops with high P inputs needs, and crops whose needs are less, which can assist in resource-efficient fertilizer plans and sustainable nutrient management for soil.

Figure 7 illustrates the potassium (K) intake of a range of crops expressed in Kg/ha [2]. The left-hand side represents crops that use the most potassium, with grapes and apple at the top and a selection of chickpea-watermelon-muskmelon afterwards and smallest amounts of K needed at the right-hand section. Again, mothbeans needed the least

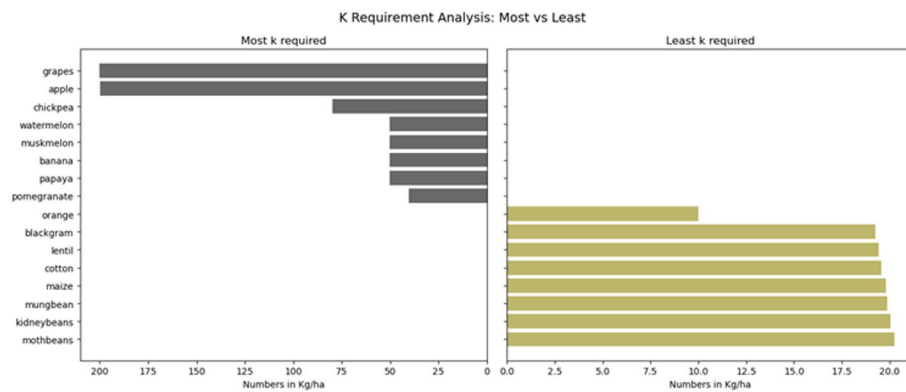


Fig. 7 Analysis regarding most and least potassium-requiring crops

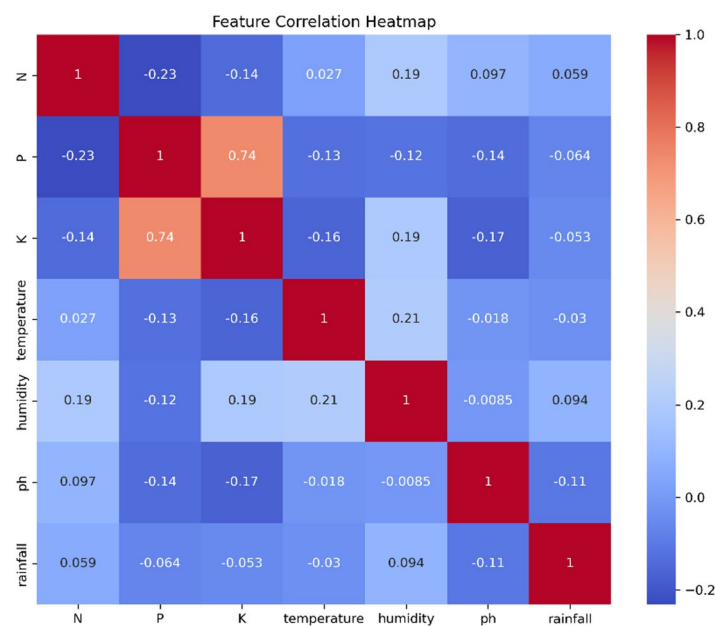


Fig. 8 Correlation among the features of the crop recommendation dataset

amount, followed by kidney beans, mungbean, and maize (with orange needing the least nutrient). This information enables informed decision making when planning fertilizer, and at the same time will also help with positive nutrient use management in a more sustainable farming context.

4.4 Model evaluation metrics

In applied agriculture, a model's usefulness should not be judged only on the metric of accuracy (Fig. 8). A model may receive a high score for accuracy, while actually performing poorly in certain areas particularly if the dataset does not have a balanced distribution among the classes [1]. To perform a more complete assessment, we will evaluate our model using a variety of performance metrics—Accuracy, Precision, Recall and F1-Score—and all the metrics include values from the model's confusion matrix: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) [24].

4.4.1 Accuracy

Accuracy reflects the proportion of correct predictions made by the model. It is fine to use when the dataset is balanced,

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

4.4.2 Precision

Precision shows how many of the crops predicted as suitable were actually suitable. It is useful when false positives should be minimized.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

4.4.3 Recall

Recall measures how well the model is able to find all of the truly suitable crops. It is useful when missing a good recommendation for a crop would be costly.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

4.4.4 F1-Score

The F1-Score takes both precision and recall and reports them as one value. It is helpful when both false positives and false negatives matter.

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

4.5 Result analysis

To assess the evaluating the performance of the proposed IoT-ML agriculture system we applied and evaluated several ensemble classifiers and assessed their performance based important metrics for evaluating—Accuracy, Precision, Recall, and F1-Score—to evaluate the potential of agricultural predictions and recommendation tasks such as crop recommendation and fertilizer recommendation. The performance of the Enhanced Gaussian Naive Bayes (EnhancedGNB) classifier outshined the others with an accuracy measurement of 99.55%, Precision of 99.56%, Recall of 99.55%, and F1-score features of 99.54%. This performance means we have a highly reliable model and can build a system that makes accurate and balanced predictions.

The Random Forest Classifier was closely behind the EnhancedGNB, with an accuracy measurement of 99.32%, and with the other measurements being very close, it confirmed the robustness and continuity in performance of the Random Forest based model as its own creation and evaluation. Bagging Classifier had a competitively measured performance and output with an accuracy of 99.09%, precision of 99.12%, recall of 99.09%, and F1-score of 99.09%. The other models improved upon by the first two classifiers, but still failed to match the EnhancedGNB and Random Forest Classifier [25]. The Gradient Boosting models report that the overall measurement reports were low across the board (Accuracy: 98.18%, F1-Score: 98.19%), they show us that there were some potential limitations with the model's ability to handle this specific dataset. Overall, these results confirmed that the EnhancedGNB classifier and its evaluation achieved the most precise measurements made in predictions, to elevated precise boundaries called “guide points”.

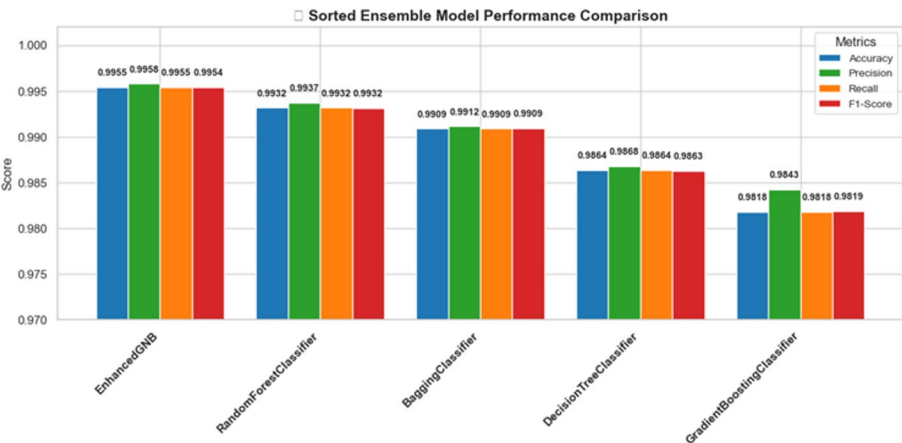


Fig. 9 Comparative analysis with different predictive models

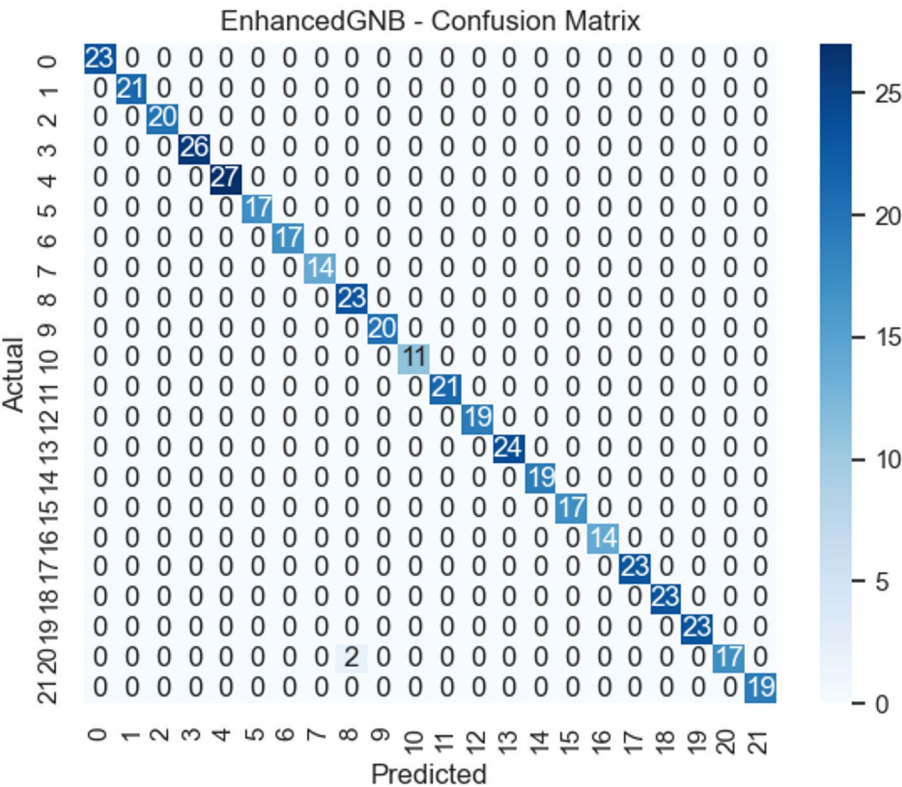


Fig. 10 EnhancedGNB correlation matrix

Therefore, we considered the EnhancedGNB the most suitable approach when implementing our proposed agricultural system (Figs. 9 and 10).

4.5.1 ROC curve analysis

The classification performance of the EnhancedGNB model across 22 crop classes was evaluated using the ROC (Receiver Operating Characteristic) curve. As shown in Fig. 11, every class achieved an AUC (Area Under Curve) of 1.00, which suggests perfect discrimination between classes. For all crops, the ROC curve closely hugs the top-left corner indicating high sensitivity and specificity. This performance provides additional

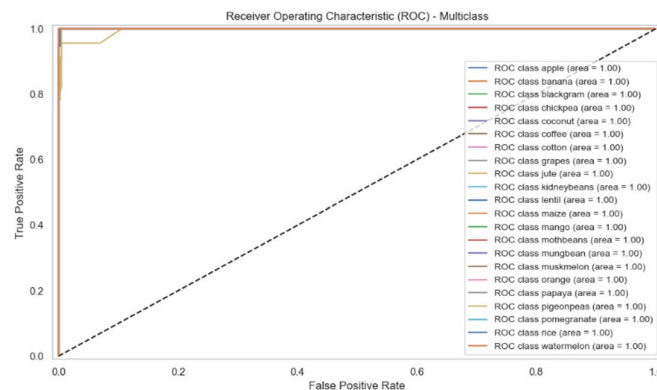


Fig. 11 Analysis of ROC curve

evidence that the EnhancedGNB model can deliver accurate crop recommendations and it can be confidently adopted for real-time operation within smart agriculture systems.

4.5.2 Crop recommendation

In our agricultural crop recommendation task, we have evaluated the predictive performance of several different ensemble models. Among the multiple models, EnhancedGNB classifier provides the best results for crop recommendations, indicating it is appropriate for real-field use. The high accuracy and balanced score of the precision, recall, and F1, suggest that there is a strong generalization of predictive capabilities and reliability in its predictions of crop suggestions [26].

The Random Forest Classifier and Bagging Classifier also produced strong results, underlining the usefulness of ensemble learning within agriculture. Each effectively identified and suggested recommended varieties of crops based on the environmental and soil parameters obtained from the deployed IoT sensors. With marginal differences amongst the other two models and EnhancedGNB, was selected for deployment, because it performed consistently better on all evaluation metrics. The predictive accuracy of the system will ensure optimal crop suggestions based on the conditions of their fields, ultimately leading to improved productivity and the use of resources [27].

4.6 Proposed IoT-ML platform

The IoT-ML-enabled agriculture platform combines smart agricultural technologies with real-time crop recommendation. The central user interface in this system is web application that is very user-friendly. When the farmer logs in to the application, it retrieves the live sensor data (e.g., soil moisture, temperature, nutrient) from the Blynk IoT cloud [10].

The sensor data is originally collected from the field using an Arduino Nano with sensors for the soil and temperature. The Arduino reads the soil moisture and temperature from its respective sensor, and the readings are sent wirelessly with nRF24L01 modules to the ESP32 microcontroller and uploaded to the cloud. Once the web application retrieves the real-time data, it quickly sends it to the deployed Enhanced Gaussian Naive Bayes (EnhancedGNB) machine learning model. The model uses the input and computes the most suitable crop recommendations specific to where the farmer is located relative

to existing environmental/soil conditions. The output is presented on the dashboard in the web application as well as in the Blynk mobile app [2, 12, 28].

The system provides a quick, reliable, and affordable avenue to contemporary agriculture. This includes real-time agriculture, precision agriculture, and data-supported decision making. By embedding the ML model into the live feedback loop, it provides farmers with immediate and accurate recommendations, distinguishing legitimate cost-effective smart growth possibilities that are scalable, reproducible and implementable under real conditions [2].

4.7 Ablation study and impact analysis

An ablation study allowed for the evaluation of the effects of removing or changing parts of the IoT–ML framework individually to determine how each of the component systems and methodological improvements to them contributes to crop recommendation performance. This study looks at how removing or modifying one system component can remove or modify (theoretical) impacts of the components on crop recommendation performance through real-time sensing, pre-processing of data, selection of model, and overall integration of disparate systems into one system. The baseline model was a Gaussian Naive Bayes classifier trained on crop recommendation data, without the use of feature standardization or variance smoothing. Under these conditions, this model yielded lower predictive performance due to being very sensitive to differences in scale and being uncalibrated for number stability. By applying feature standardization to the training set, the model showed improvements in predictive performance, demonstrating the importance of scaling agronomic variables that were collected in different units such as parts per million (ppm), degrees Celsius (°C), and millimeters (mm).

The Variance Smoothing was integrated into the Gaussian Naive Bayes model and further enhanced with the EGNB term (Enhanced Gaussian Naive Bayes). With this update, the likelihood of being accurate decreased numerically, allowing better estimations of accuracies, precision, recall, and F1 scores. As such, this supports the importance of enhancing the probabilistic model for reliable generalization. In comparing the EGNB with three tree-based classifiers that were preprocessed in the same manner as the EGNB, it became apparent that while Random Forest, Bagging, and Gradient Boosting produced good results, they were only slightly better than the EGNB for the increased computational load. Therefore, the enhanced version of the approached using the Probabilistic Model structure better accommodates the Agronomic Dataset structure as compared to using ensemble hierarchies.

5 Discussion

The experimentation conducted with the proposed framework for machine learning driven by the Internet of Things provides evidence that this framework is capable of providing real-time crop recommendations when applied to established agricultural conditions. The Enhanced Gaussian Naive Bayes (EGNB) classifier performed far better than any other classifier tested, by achieving an overall accuracy of 99.55% as well as maintaining high levels of precision, recall, and F1 scores in each test. The successful use of ensemble methods like Random Forest, Bagging, and Gradient Boosting in Precision Agriculture is primarily attributed to their ability to effectively learn non-linear interactions among features [29]. While all three ensemble models performed slightly

worse than the EGNB in terms of accuracy, it should be noted that model complexity does not guarantee better performance when working with structured agronomic datasets. Furthermore, since this research included soil and climate parameters such as N, P, K, temperature, humidity, pH, and precipitation that possess statistically stable distributions and limited inter-feature dependency, these same conditions enabled the EGNB to perform as competitively as other probabilistic classifiers.

The performance of the EGNB model can be attributed to its probabilistic formulation combined with feature standardization and variance smoothing, which has the effect of stabilizing likelihood estimation across heterogeneous variables measured on different physical characteristics and different value ranges. Ensemble models may introduce additional variance when the true relationship among features is sufficiently captured by more basic statistical assumptions [30]. Similar experiences have been reported in other crop recommendation systems, where the level of accuracy achieved by Naive Bayes- and lightweight probabilistic-based systems had comparable levels of accuracy when trained on highly curated agricultural datasets. However, many other such earlier systems reported a predictive accuracy of approximately 90–97% and were frequently limited to offline analysis or simulation environments. In addition to the EGNB-based system operating at levels of performance greater than those reported in prior systems, the EGNB-based system will also operate in a real-time IoT-enabled platform. The High accuracy of the model can be attributed to a balanced precision and recall, meaning that the system is very successful at reducing false positive (recommendation of unsuitable crops) and false negative (omission of suitable crops) rates [31]. This balance is essential in providing reliable decision support for operational agriculture due to the rapidly changing conditions of agriculture and the extensive use of real-time sensor data. The Results of the Receiver Operating Characteristic analysis also indicate that the majority of classes had very high Area under the Curve (AUC) values, demonstrating that there is excellent separation between each class and that class imbalance or data leakage does not significantly influence the performance of the model [32].

A growing number of studies are documenting that the use of precision agriculture has increased dramatically over the last several years. The majority of studies to date have focused on developing specific components of precision agriculture as opposed to providing a comprehensive system-level solution. The majority of studies to date have developed machine-learning-based crop recommendations without using live field data and instead relied on static historical soil and climatic data [33]. A substantial body of literature also exists that uses IoT devices to monitor crops via sensor data and provides graphical visualizations of the data collected. However, the development of IoT devices has not included the incorporation of intelligent analytics to convert the collected sensor data into actionable crop recommendations. Therefore, there remains an extensive gap in research because there is currently no comprehensive closed-loop IoT-ML system that provides continuous site data monitoring; generates live decision-making through live inference; and provides context-aware recommendations based on the best farming practices [34]. The majority of existing systems have been tested and evaluated either in a simulated or laboratory testing environment, creating an incomplete assessment of each system's practical application constraints, including cost, sustainability, scalable use, energy efficiency, and farmer usability. As well, no complete, end-to-end system has been developed that integrates real-time sensing capabilities with intelligent

machine-learning-based decision-making and provides low-cost, operationally viable deployment platforms for farmers, particularly smallholder farmers in developing countries [35]. These factors lead to an urgent need for a complete, comprehensive, low-cost, and operationally viable precision agriculture solution that directly integrates real-time sensing capabilities with intelligent decision-making capabilities for immediate implementation [36–40].

5.1 Limitations

Nevertheless, while our IoT-ML Framework has produced excellent predictive results to this point in time, we also want to address several limitations of this initial study. To begin with, although we collected soil moisture and temperature directly from working sensors in real-time, the NPK values that were used to build and test our model were generated by simulating NPK values from thousands of real-world agronomic data samples rather than being collected through a fully calibrated in-field nutrient sensor network. This was done to facilitate experimentation under controlled conditions and because it was necessary for us to assemble an entire data set in order to test our framework. Therefore, although we have produced results that appear to be accurate based upon our simulations, we cannot expect the same accuracy under actual agricultural growing conditions, which have a much higher level of variability in terms of the spatio-temporal nature of nutrient cycling. Our intent is to incorporate calibrated nutrient sensors and implement a systematic sensor validation against laboratory-based soil nutrient interpretation in subsequent work in order to improve the overall quality of nutrient measurements.

6 Conclusion and future scope

The IoT-ML enabled farm system as designed was successfully developed, integrating wireless sensor networks with cloud computing and machine learning to provide real time crop suggestions. The system made use of an Arduino Nano to sense soil moisture and temperature, with data sent over the nRF24L01 module to the ESP32 gateway. Data was uploaded to the Blynk IoT cloud from the ESP32 gateway, which the web application used in conjunction with the Enhanced Gaussian Naive Bayes (EnhancedGNB) model, would process and send real-time crop suggestions using a web dashboard to the agriculturalist.

The system offers the farmer an easy-to-use platform to receive location-specific crop suggestion capabilities utilizing a web dashboard and collating real-time data. The capabilities of the receiver and transmitter circuits in terms of hardware integration confirm a successful communication between low-cost microcontrollers and sensors, confirming a viable, scalable and cost-effective solution suitable for rural deployment discussed in the results and discussions sections of this thesis. Residential grouped data suggests the EnhancedGNB model to be robust with an ability to accurately classify crops based on specific environmental conditions, with an AUC rating of 1.0 for all classes in the ROC curve results. The research presented in this study provides a low-cost, real time IoT-ML-enabled crop recommendation system for precision agriculture. Specifically, the system is built using an Arduino Nano with sensors for soil moisture, temperature, and simulated NPK. Sensor data is wirelessly transmitted to the ESP32 gateway via nRF24L01 modules. The ESP32 then sends the data to the Blynk IoT cloud from which

a Blynk web application retrieves the data and the deployed Enhanced Gaussian Naive Bayes (EnhancedGNB) model processes it.

The proposed system achieved an incredible accuracy of 99.55% in crop recommendations, thus providing very reliable results to end users. The platform allows for real-time environmental monitoring and real-time crop recommendations via a web-based dashboard optimized for mobile platforms and the Blynk mobile application [37].

The specifications of the proposed system use low power consumption, are low cost, and integrates the data collection system into machine learning algorithm in a way that is most appropriate for the context of Indian agriculture, where it will be potentially beneficial for remote or underdeveloped places [38]. By enabling informed, data-driven, field-ready farming decisions, this platform is well-positioned to contribute positively to crop yields, resource efficiency, and sustainability in agriculture.

For future studies, the use of new environmental values and variables such as soil pH, amounts of rainfall, and exposure to sunlight are the focus of the variables that can be added to improve the system. In addition, intelligent deep learning using models and edge computing on the ESP32 could mean the inferencing can occur on-device, which would reduce latency in the application. Continued empirical field tests in different agricultural regions and in different crop cycles, conditions, and settings would give further evaluation and implication of the model.

Author contributions

N.R.S helped in methodology, visualization, data curation, formal analysis and writing—original draft. A.K worked in conceptualization, investigation, and writing—review and editing. S.P helped in investigation and writing—review and editing. K.K worked in supervision and writing—review and editing.

Funding

Open access funding provided by Symbiosis International (Deemed University). This research is supported by Research Support Fund of Symbiosis International (Deemed University).

Data availability

All data generated or analyzed during this study are available from the corresponding authors upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 8 October 2025 / Accepted: 19 January 2026

Published online: 06 February 2026

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